

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Humanities & Social Sciences

2. Subject Name: AI / ML Laboratory **L-T-P:** 0-0-3 **Credits:** 2

3. Pre-requisites: AI/ML Theory

4. Syllabus and reference books:

Syllabus:

1. Shortest Path Algorithms (Dijkstra & Bellman Ford) [Week - 1]
2. A* search, uniform cost search [Week - 2]
3. BFS & DFS [Week - 2]
4. Game Trees [Week -3 & Week - 4]
5. Decision trees [Week - 5]
6. Bayesian Network & Applications of Finance [Week - 6 & Week - 7]
7. Gradient descent [Week - 8]
8. Perceptron [Week - 8]
9. Deep neural networks [Week - 9]
10. Back propagation [Week - 9 & Week - 10]
11. Reinforcement Learning [Week - 11 & Week - 12]

Reference Books:

1. Artificial Intelligence - Modern Approach (3rd Edition) - Stuart Russell & Peter Norvig
2. Reinforcement Learning in Finance - Ashwin Rao

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures (3 - hour lecture, 1 per week)
1.	Shortest Path Algorithms (Dijkstra & Bellman Ford)	1
2.	A* search, uniform cost search	1
3.	BFS & DFS	1
4.	Game Trees	1
5	Decision trees	1
6	Bayesian Network & Applications of Finance	2
7	Gradient descent	1
8	Perceptron	1
9	Deep neural networks	1
10	Back propagation	1
11	Reinforcement Learning	2
Total number of hours		40